

What is map ?

Data structure that store value pairing with its key

Terms

load factor: The amount of data points vs the amount of storage (`data.len / storage.capacity`)

key: a value that is hashable and is used to look up data. The hash has to be consistent.

value: a value that is associated with a key

collision: when 2 keys map to the same cell.

What is the load factor in the context of maps?

The load factor is the ratio of the number of data points to the amount of available storage. For example, if seven items are in a map with ten available storage units, the load factor would be 0.7.

What is a collision in a map?

A collision occurs when two different keys map to the same storage cell or bucket in the underlying data structure.

What are the key requirements for a hashing function?

A hashing function must produce a consistent hash, always returning the same output for a given input, and must convert the key into a numeric value that can be used to determine the storage location.

What is the recommended ideal load factor for a map?

The ideal load factor is approximately 0.7, which helps minimize collisions and maintain efficient storage and retrieval operations

Why do maps typically have $O(1)$ constant lookup time?

Maps have $O(1)$ lookup time because hashing functions convert keys to storage indices quickly, and assuming a good hash distribution with minimal collisions, accessing and retrieving a value takes constant time.